How-To Library

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Configuring Disaster Recovery for Streaming Sources in Big Data Streaming 10.2.2

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Abstract

You can use a streaming mapping to read from streaming sources, such as Apache Kafka. This article describes about configuring disaster recovery for the streaming sources to restore the processing after a mapping failure.

Supported Versions

- Big Data Streaming 10.2.2

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Overview

You can create streaming mappings to stream machine, device, and social media data in the form of messages. You can use a Messaging connection to get data from streaming sources, such as Apache Kafka. The Spark engine runs a streaming mapping continuously.

When a streaming mapping that uses a streaming source fails, by default, the Spark engine does not continue processing data from where it stopped. When the streaming mapping restarts, the Spark engine processes data based on the start position offset value which results in data duplication or data loss.

To achieve data continuity and avoid duplication of the streaming process, you can configure disaster recovery for the streaming sources. With disaster recovery, you can resume the streaming process from the point of failure and minimize the impact on business.

To configure disaster recovery for a streaming source, perform the following tasks:

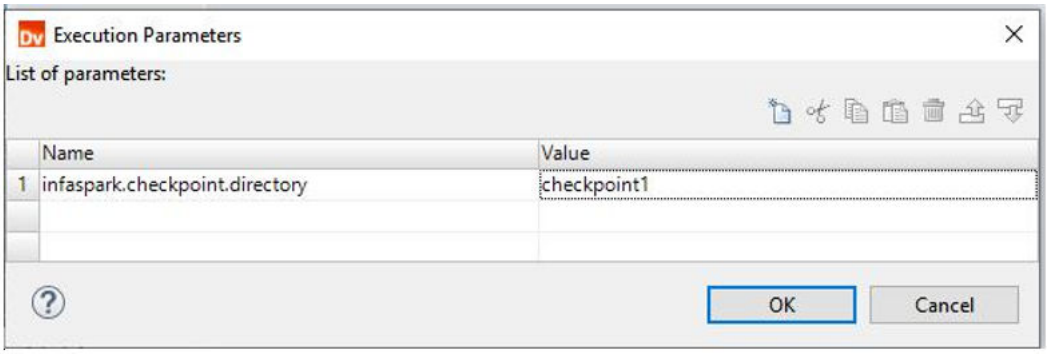
- Configure a checkpoint directory in the Spark Execution Parameters.
- Configure a state store directory in the mapping run-time properties or the execution options for the Data Integration Service.

Configuring Checkpoint

Checkpoint saves the state of the streaming process in a highly available HDFS storage so that it can be recovered upon failure. In case of a mapping failure, the Spark engine uses checkpoint to start reading data from the point of failure when the mapping restarts.

To enable checkpoint in a streaming mapping, configure the `infaspark.checkpoint.directory` property. The directory specified in the `infaspark.checkpoint.directory` property is created in the state store directory. Ensure that the checkpoint directory path does not start with a slash (/). You can configure the checkpoint directory through the Spark Execution Parameters.

The following image shows the `infaspark.checkpoint.directory` property that you can configure through the Spark Execution Parameters:

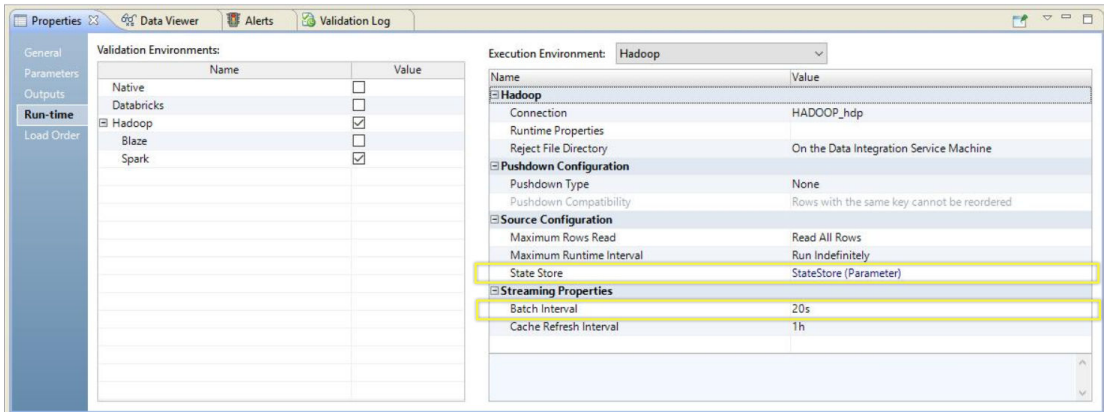


Configuring State Store

In a streaming mapping, state store saves the checkpoint information based on the batch interval during run time. Batch interval is the amount of time after which a batch is submitted for processing. You can specify a value for the batch interval through the mapping run-time properties. When a mapping fails, the Spark engine restarts processing by reading the information stored in the state store directory.

Specify the HDFS location on the cluster to store information about the state of the Spark job. Default is `/tmp`. You can configure the state store as part of the configuration of execution options for the Data Integration Service. You can also configure the state store through the mapping run-time properties.

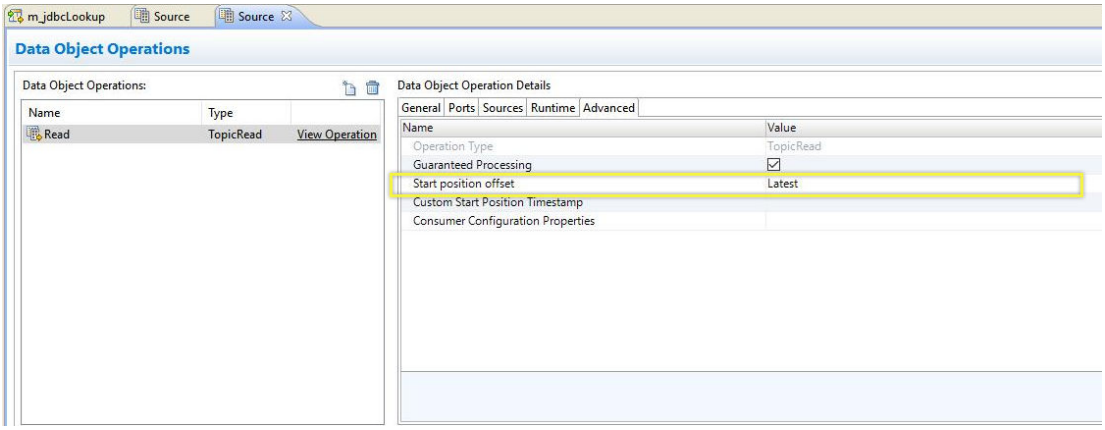
The following image shows the **State Store** and **Batch Interval** properties in the mapping Run-time properties:



Configuring Offset

When you start a streaming mapping, by default, the Spark engine processes data based on the start position offset value. Start position offset is the point from which a streaming source starts reading messages. You can configure the start position offset in the **Data Object Operations** panel.

The following image shows the **Data Object Operations** panel:



You can select one of the following options:

- Custom. Reads messages from a specific time.
- Earliest. Reads the earliest messages available.
- Latest. Reads messages received after a mapping has been deployed.

Note: Start position offset does not have any effect on the disaster recovery process.

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